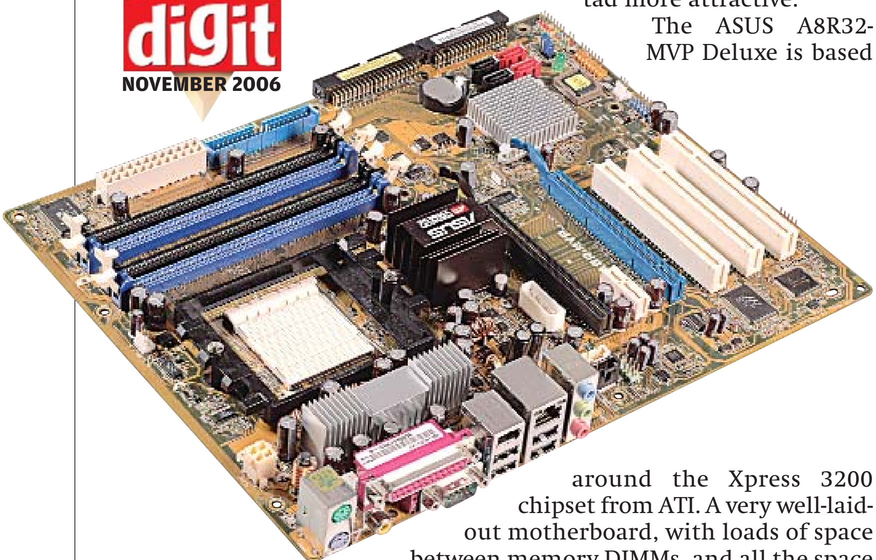
3200. These chipsets support CrossFire (ATI's multi-graphics counter to NVIDIA's SLI). With the gaining popularity of AM2, even high-end Socket 939 motherboards such as these have had price slashes; making even these ageing DDR-based solutions seem a tad more attractive.



ASUS A8R-MVP
ASUS' Budget CrossFire offering

The ASUS A8R32-MVP Deluxe is based around the Xpress 3200 chipset from ATI. A very well-laid-out motherboard, with loads of space between memory DIMMs, and all the space you require to install a large heatsink-fan. It's feature rich as well—the two PCI-Express slots are capable of simultaneous x16 operation. There is sufficient spacing between the two PCI-E slots to accommodate even the largest of double-slot graphics cards.

The board makes quite a strong point for itself as a quality performance product, but isn't future-proof due to the processors and memory for this platform becoming extinct. There's no nifty chipset heatsink on this one, and unlike the nForce 4 chipset, which was a single chip solution, the Xpress 3200 solution on the A8R32-MVP Deluxe bears a Northbridge and Southbridge chip.

The A8R32-MVP Deluxe is an overclocker's delight, and ASUS advertises this right on the box. The BIOS options are numerous, and the HT (HyperTransport) speed can be bumped up to insane levels—we tried overclocking our board and were able to get a very respectable 600 MHz boost to a 1.8 GHz AMD 64 3000+ processor. The board supports aggressive CPU and memory voltage increases, which is necessary to overcome the limitations of overclocking on stock voltages.

The ASUS A8R-MVP is the smaller brother of the Deluxe version above, and just a peek at both the boards will tell you why. The layout is decent for a single graphics solution, but using this board for a CrossFire setup could pose a problem due to insufficient spacing between the graphics slots. The build quality of the components on the A8R-MVP is good, but again, the departure from the Deluxe version is apparent.

We received two motherboards based on ATI's RS 482 chipset, one each from Sapphire and Axper. Sapphire, best known for their Radeon-based graphics cards, are somewhat new to the Indian market, while Axper is a new entrant into the motherboard arena, and this is the first time we've received motherboards from them. The RS 482 chipset features a decent onboard graphics solution—ATi's X300, which is a Direct X 9.0 compliant solution. The onboard graphics solution will share memory (up to 256 megabytes of it), and is not really recommended for gaming.

Both these boards support dual-channel memory; however, you just get two DIMMs, unlike the other boards which sport four. Both sport rudimentary overclocking in their respective BIOSes, although we found the Sapphire board a bit more responsive to tweaks. In terms

The Memory Question

While deciding on a motherboard, there is the perennial headache of choosing matching memory. Of course, much of your memory decision rests on choice of socket, and DDR will work fine with Socket 939 AMD processors, while AM2 processors and all Intel's 775 LGA CPUs demand DDR2. However even in the two classes—DDR and DDR2—you've got a lot of choosing to do!

Intel processors (Pentium 4 / D / Core 2 Duo) are basically memory bandwidth sensitive (read: MHz) due to their architectures. Look for memory with higher speeds rather than paying more for low latencies.

So what would be good memory for a Core 2 Duo? Practically, a processor cannot demand more data from memory than what its FSB bandwidth allows. This holds true for Intel processors only (AMD 64s and their ilk have the memory controller integrated on the CPU die, so the concept of FSB is absent). Figuratively, for an FSB of 1066 MHz, the bandwidth would be (1066 x 32 bits)/8 = 4264 MBps. If we consider that memory should be twice as fast (due to latencies involved), DDR2 667 would offer a bandwidth of (667 x 64)/8 = 5336 MBps. In case of dual-channel, this would be (5336 x 2 = 10672 MBps), or roughly two and a half times higher than the FSB bandwidth. Faster memory is available (DDR2 800 and DDR2 1000 MHz), but it seems all that additional bandwidth isn't doing much more except costing more. We recommend you stick with DDR2 667 for now for any Intel-based rig, go for more memory, but not faster memory, if you need more performance. Opt for higher speeds only if you plan to do a lot of aggressive overclocking—say 3.2 GHz and above for a Core 2 Duo processor.

With AMD processors, their integrated memory controller means that not only do they scale better with speed (MHz) increments than their Intel counterparts, they show steeper performance increments with lower-latency memory. Kind of a double-edged sword considering prices, and the fact that performance-wise, speed and latency are inversely proportional. But bear in mind that AMD processors are much more sensitive to both latency and bandwidth, so you'll need faster memory here.

Socket 939 performs optimally with DDR 400 MHz memory; try and get as low a latency as your budget allows. For AM2-based processors, a general rule of thumb is to balance out the speed and latency of the RAM you're buying, else your bill will skyrocket. Opt for DDR2 667 MHz memory in the latency range of 4-4-4-10. While lower latencies are possible, as are higher clocked parts, the performance increment doesn't justify the extra you'll shell out. For example, 2 GB of low-latency DDR2 800 MHz memory will cost you in excess of Rs 16,000. DDR2 800 MHz only makes sense if you are pairing it with a fast processor, say a 2.4 GHz dual-core AMD 64 at the very least.